Tutorial 2: Worked solutions

Electrical Engineering I · Module 92315 · Tutorial 2

From Fundamentals to AI Apocalypse · personal study guide

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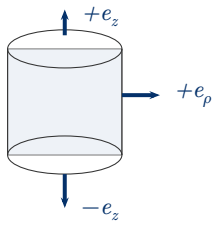
These solutions build total charge by integrating the volume charge density over the region where it exists. Track the coordinate-dependent volume element and the density's support.

Task 4: Cylinder coordinates

The space charge density is constant, $\varrho_V = \varrho_0$, in the hollow cylindrical region $a \le \rho \le b$, $0 \le \varphi \le 2\pi$, $0 \le z \le c$, and zero outside. Assume $0 \le a < b$ and $c > 0$.

Connected topic: Flux and Gauss's law (§1.10) develops cylindrical coordinates and radial geometry; the volume element supplies the appropriate shell weighting here.



Integrate only the charged hollow region. The charge occupies radii a through b and height c; its cylindrical volume element is $\rho\, d\rho\, d\varphi\, dz$.

4: Total charge in the hollow cylinder

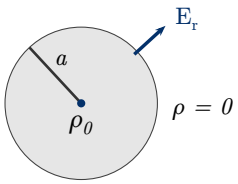
Try first: write the cylindrical volume element, then integrate across the inner and outer radii, the full angle, and the height.

Answers: see "Task 4: Cylinder coordinates" under Answers at the end.

Task 5: Spherical coordinates

The density varies radially as $\varrho_V(r) = \varrho_0 r^2/a^2$ inside a sphere of radius a and is zero for $r > a$. Assume $a > 0$.

Connected topic: Flux and Gauss's law (§1.10) uses spherical symmetry and spherical shells; here each shell contributes its density times its shell volume.



The charged sphere has radius a. Its density rises with r^2, so outer spherical shells carry more charge per unit volume than inner ones.

5: Total charge in the sphere

Try first: multiply the radial density by the spherical shell volume $4\pi r^2 dr$ and integrate from the centre to a.

Answers: see "Task 5: Spherical coordinates" under Answers at the end.

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Answers

Task 4: Cylinder coordinates

4 · Cylindrical volume integral

1. Total charge is the volume integral of volume charge density over the charged region. In cylindrical coordinates, a small volume is a wedge-shaped piece with volume $\rho\, d\rho\, d\varphi\, dz$:

$$Q = \iiint_V \varrho_V \, dV, \qquad dV = \rho \, d\rho \, d\varphi \, dz$$

2. Substitute the constant density and the stated limits. The inner radius a excludes the empty central cylinder:

$$Q = \varrho_0 \int_0^c \int_0^{2\pi} \int_a^b \rho \, d\rho \, d\varphi \, dz$$

$$Q = \varrho_0 \int_0^c \int_0^{2\pi} [(\rho^2/2)]_a^b \, d\varphi \, dz$$

3. Evaluate the radial term, then the full revolution and cylinder height:

$$Q = \varrho_0 \, ((b^2 - a^2)/2)(2\pi)c$$

$$Q = \varrho_0 \, \pi(b^2 - a^2)c$$

The sign follows the sign of ϱ_0. **Check:** the factor $\pi(b^2 - a^2)c$ is exactly the hollow cylinder's volume, and ϱ_0 times volume has units C.

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Task 5: Spherical coordinates

5 · Spherical volume integral

1. For a spherically symmetric density, integrate over concentric shells. A shell at radius r has volume $4\pi r^2 dr$, and only radii 0 through a contain charge:

$$Q = \iiint_V \varrho_V \, dV = 4\pi \int_0^a \varrho_V(r) \, r^2 \, dr$$

2. Insert the given density. One factor r^2 comes from the shell area and another from ϱ_V:

$$Q = 4\pi \int_0^a (\varrho_0 r^2/a^2) \, r^2 \, dr$$

$$Q = (4\pi\varrho_0/a^2) \int_0^a r^4 \, dr$$

3. Integrate r^4 and apply both limits:

$$Q = (4\pi\varrho_0/a^2)[r^5/5]_0^a$$

$$Q = 4\pi\varrho_0 a^3/5$$

The sign follows ϱ_0. **Check:** the average density is $3\varrho_0/5$, giving $(3\varrho_0/5)(4\pi a^3/3)$, the same result.

Made by Thanate A

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